

Smaran Pokharel

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Education

Dakota State University

Madison, SD

B.S. Computer Science (AI Specialization), Minor in Mathematics

Expected May 2028

GPA: 4.0 | President's Academic Honors List | DSU Rising & Champion Scholarship

Relevant Coursework: Object-Oriented Design, Software Security, Advanced Data Structures, Parallel Programming

Competitions & Achievements

eCitadel – Database & Router

Participant, Jun 2026

Defended a live Debian/MariaDB server against continuous Red Team attacks in a national live-fire cyber defense exercise; hardened MariaDB configuration, enforced UFW firewall rules, removed unauthorized users, and maintained 100% service uptime throughout the event.

HackDavis 2026 – AI / Backend Developer

Best Statistical Model, May 2026

Won Best Statistical Model out of 800+ participants by building a grocery price forecasting platform leveraging 25 years of USDA data and Google TimesFM 2.0 via BigQuery ML, achieving R^2 of 0.91 to deliver actionable buy/wait recommendations for SNAP-eligible households.

HiveCTF – Competitor

1st Place, Apr 2026

Ranked 1st within the DSU bracket in an open Capture the Flag competition, solving challenges across web exploitation, cryptography, and reverse engineering under time pressure.

Amazon Nova AI Hackathon – Backend Developer

Participant, Mar 2026

Engineered the FastAPI simulation backend for GeoStrategic Intelligence Sandbox — a turn-based geopolitical engine where Amazon Nova Lite AI agents autonomously control country blocs, dynamically adjusting stability, military spending, and diplomatic relations in response to user-triggered world events.

NCAE Cyber Games Regionals 2026 – Database & Router

5th Place, Feb 2026

Placed 5th regionally in a simulated enterprise network defense competition; maintained critical database and routing services under sustained adversarial pressure while monitoring traffic and mitigating live attacks.

Cybersports LTD CTF Summit – Competitor

4th Place, Oct 2025

Placed 4th in a competitive timed CTF, solving challenges across web security, reverse engineering, and digital forensics — demonstrating applied offensive security skills under pressure.

Projects

Fipher Keys

React, TypeScript, Vite, Supabase, ExcelJS, Tesseract OCR | [github/Fipher-Keys](#)

Aug 2026 – Present

- Built a full-stack key management system for DSU Facilities Management which helps track around 1,000+ keys spanning 28+ years of records across 400+ holders, replacing a hand-maintained spreadsheet with a campus map, dashboard, directory, and catalog views backed by a swappable localStorage/Supabase data layer with row-level security and multi-organization support.
- Implemented Excel round-tripping via ExcelJS that imports legacy multi-sheet key spreadsheets (matching columns by heading across six source sheets) and exports back to a workbook matching the original formatting, plus PDF request/return form parsing with OCR fallback via pdf.js and Tesseract.
- Designed the issue/return workflow, person and key detail pages with cross-linked history, and a platform admin console for managing organizations and access codes.

Terminal Quest

React, TypeScript, Vite, Supabase, Gemini API | [github/terminalquest](#)

Apr 2026 – Present

- Built a full-stack browser-based dungeon RPG where Linux commands (ls, cd, cat, find, mv, mkdir) drive real in-game movement, item discovery, locked doors, and room transitions — mapping filesystem concepts directly to dungeon mechanics.
- Integrated Gemini AI via Supabase Edge Functions for adaptive level generation, procedural room creation, in-game tutor responses, and run reports, with deterministic local fallbacks ensuring full playability when the API is unavailable.
- Implemented Supabase Auth (email + Google OAuth), persistent quest saves, per-player stats tracking (commands used, keys found, levels completed), achievement unlocks, and a command mastery profile page.

EveryCent

React, TypeScript, Python, BigQuery ML, Claude API, TimesFM 2.0 | [github/EveryCent](#)

May 2026 – Aug 2026

- Built a grocery price forecasting platform using 25 years of USDA retail price data and Google TimesFM 2.0 via BigQuery ML ALFORECAST, achieving an R^2 of 0.91 vs 0.76 for a moving-average baseline (MAE: \$0.06).
- Delivered buy/wait/stable recommendations for essential grocery items with SNAP cycle alignment — giving low-income families access to the same forecasting intelligence used by retailers.
- Integrated Claude Haiku with live web search to generate real-time market context explaining why prices are moving, surfaced alongside TimesFM forecast charts and a smart shopping list.

HandAll

- React, TypeScript, FastAPI, LangGraph, OpenAI, Google Calendar, Docker | [github/HandAll](#) Jan 2026 – Mar 2026
- Architected a full-stack AI student planner with a React/TypeScript frontend, Node.js/Express backend, and a Python FastAPI service running LangGraph agents for schedule decomposition, burnout-aware task distribution, and motivation-driven replanning.
 - Built Google Calendar OAuth integration and iCal/ICS import with a preview-before-sync workflow; implemented a motivation slider that triggers live AI schedule rebalancing between recovery and lock-in modes.
 - Contributed the Google Calendar connect flow and product integration as part of a 4-person team, delivering the full product in a 36-hour hackathon sprint.

Dr. Prompt

- JavaScript, Chrome Extension API, Claude API, Manifest V3 | [github/DrPrompt](#) May 2026 – Jun 2026
- Built a Manifest V3 Chrome extension that rewrites rough AI prompts into clearer, more effective versions using the Claude API, with a local rule-based fallback optimizer when the API is unavailable.
 - Injected inline prompt suggestion bubbles across five AI platforms (ChatGPT, Claude, Gemini, Mistral, Perplexity) and added a right-click context menu to send selected webpage text directly into the optimizer.
 - Shipped a searchable Popular Prompts library with category filtering and before/after token count comparison to help users understand prompt efficiency.

Experience & Leadership

Peer Tutor

Madison, SD

- Trojans Tutoring – Dakota State University Jan 2025 – Present
- Delivered 45+ one-on-one tutoring sessions for 30+ students across various computer science and mathematic courses, adapting explanations to individual learning gaps and pacing.
 - Translated complex technical concepts into approachable frameworks, consistently improving student comprehension and assessment performance.

Front Desk Officer

Madison, SD

- Facilities Management – Dakota State University Apr 2026 – Present
- Maintain and update a keys database for campus facilities, manage fleet operations using AG Fleet Commander, and process work orders and inventory updates to support day-to-day facilities workflows.
 - Serve as a primary point of contact for service coordination requests, ensuring accurate recordkeeping and timely resolution of facilities management tasks.

Treasurer

Madison, SD

- DSU International Club Aug 2024 – May 2026
- Manage a \$4,000 annual budget to plan and fund events for 250+ members, overseeing expense tracking, fund allocation, and financial reporting to ensure accountability and efficient use of club resources.

Technical Skills

- Languages:** JavaScript/TypeScript, Python, C/C++, C#, SQL, Bash
- Frontend:** React, Vite, Tailwind CSS, HTML/CSS
- Backend:** Node.js, Express, FastAPI, REST APIs
- Databases:** PostgreSQL, SQLite, Supabase
- AI/ML:** OpenAI API, Claude API, LangGraph, LLM Integration, Google TimesFM
- Security:** Linux hardening, UFW, MariaDB security, CTF (web exploitation, cryptography, reverse engineering)
- Tools:** Git, GitHub, Docker, Postman, VS Code, Playwright
- Systems:** Linux/Unix, CLI Tools